

Autopan

An LFO auto-panner and tremolo, free or locked to the tempo.



Overview

Autopan moves your sound with an LFO. In PAN mode it sweeps the sound across the stereo field; in TREM mode both sides move together, so the level pulses up and down - a tremolo.

The LFO can run freely in Hz or lock to the project tempo, in four shapes from a smooth sine to stepped random. A gentle drift can make the rate, depth and centre wander slowly, so the movement never feels mechanical.

Reach for it for

- Slow, wide movement on pads, keys and guitars
- Tempo-synced ping-pong panning
- Classic amp-style tremolo
- Choppy, rhythmic gating effects (square wave)
- Random, wandering stereo motion

Controls

Mode (header)

PAN	Moves the sound from side to side across the stereo field.
TREM	Tremolo: both sides move together, so the level rises and falls. At 100% DEPTH the sound dips to silence at the bottom of each cycle.

Display

Pan field / TREM GAIN	In PAN mode, a line from L to R with the range the sweep covers and a dot where the sound is now. In TREM mode, a bar showing the current gain.
LFO scope	The LFO shape with a marker riding on it. With RND, a running trace of the random movement.
INPUT	A VU meter of the input level (0 VU = -18 dBFS average).

Autopan

SINE / TRI / SQR / RND	The LFO shape. Sine and triangle give smooth sweeps; square jumps between two positions; random moves to a new place each cycle.
MONO / STEREO	PAN mode only. MONO sums the input to mono and pans it with an equal-power law (unity in the centre) - a true pan. STEREO works as a balance control and keeps the stereo image: the sweep fades the opposite side down.
FREE / SYNC / FLOW	How the rate is set: FREE in Hz, SYNC locked to the project tempo and timeline, FLOW tempo-synced but running continuously through loops.
RATE	The LFO speed: 0.05 - 20 Hz in FREE, or a note value from 4 bars to 1/32 (with dotted and triplet values) in SYNC and FLOW.

SMOOTH	Rounds off the square wave's corners and turns the random shape from stepped jumps into glides, 0 - 100%.
PHASE	Shifts the LFO within its cycle, 0 - 360 degrees. In SYNC, use it to move the sweep against the beat.
DEPTH	How far the sweep travels (PAN) or how deep the level dips (TREM), 0 - 100%.
CENTRE	PAN mode: where the sweep is centred, from -100% (left) to +100% (right).

Drift

A slow random wander that keeps the movement from sounding mechanical.

DRIFT	One dial, 0 - 100%. It makes the LFO rate and the depth wander slowly, and in some presets the pan centre too (in SYNC the rate stays locked on average and the timing wobbles instead). 50% is the standard amount, 100% twice that, 0 none; the wander also gets a little faster as you turn it up.
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IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED at the knob's upper right is dark when GUARD is off, red when it is on, and glows brighter while it is catching peaks.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

OFF / CLIP / LIMIT

The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

Master

MIX

Dry / effect balance, 0 - 100%. Lower settings make the movement subtler.

OUTPUT

Output level, -24 to +12 dB.

OUT meters

Output level, left and right, with a floating peak. Click a meter to clear the peak.

The header

PRESETS, arrows and display

PRESETS (or a click on the display) opens the list of factory presets; the ◀ / ▶ keys step through them. The display shows the preset number and name.

Theme picker

Changes the panel's colour theme.

S / M / L

Panel size: small, medium or large. The panel also scales to fit the plugin window.

?

Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

HW

Switches between the hardware-style panel and the flat look.

The presets set the mode, shape, smoothing, rate and the depth, centre and drift amounts; phase, the MONO / STEREO input, mix, output and the IN / OUT strip are left as they are. The last entry in the list resets everything to the defaults.

Presets

Init	The default settings: a free-running sine pan at 0.12 Hz, 30% depth, with a little drift. A new instance opens on it.
Gentle Sway	Sine pan over one bar at 50% depth, with a little drift.
Classic Auto-Pan	Full-width sine pan on quarter notes.
Hard Ping-Pong	Square wave, full depth, on eighth notes: the sound jumps hard left and right.
Wandering	Smooth random pan over two bars with plenty of drift.
Slow Swirl	Free-running sine pan at 0.8 Hz, with the rate drifting.
Amp Trem	Sine tremolo at 5 Hz, 60% depth.
Choppy Trem	Square tremolo on 1/16 notes at full depth - a rhythmic chop.
Warble Trem	Triangle tremolo on eighth-note triplets, 50% depth, with rate and depth drift.
Reset all to defaults	Every control back to its starting value.

Tips

- Keep a stereo source's image intact with STEREO; use MONO when you want the whole sound to travel from speaker to speaker.
- Raise SMOOTH on a square-wave pan to soften the clicks of the jumps.
- In SYNC, PHASE lets you make the sound arrive left or right exactly on the downbeat.
- For tremolo that sits in a mix, try DEPTH around 30 - 60% rather than full.

File: gfx-autopan.jsfx